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Course code:CSA0712

Course title: COMPUTER NETWORKS

CO4-AT1- Coding challenge

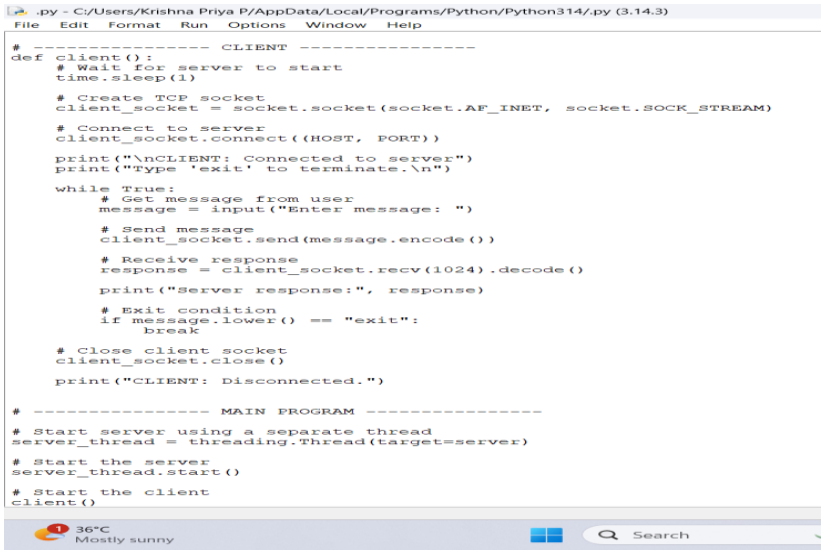
Question 1 — TCP Client–Server Application

Aim:

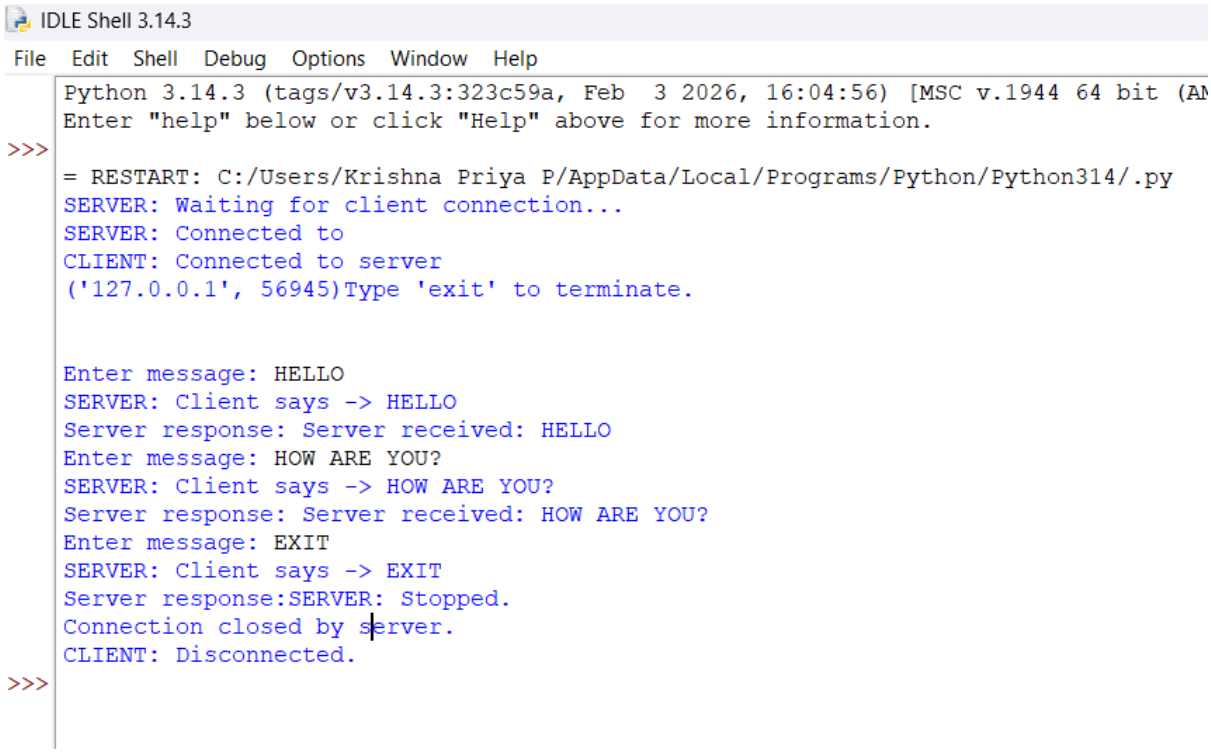
To develop and implement a TCP Client–Server application using Python, where the server and client communicate through TCP sockets to send and receive multiple messages, and terminate the connection when the client sends "exit".

CODING:




.py - C:/Users/Krishna Priya P/AppData/Local/Programs/Python/Python314/.py (3.14.3)
File Edit Format Run Options Window Help
---------- CLIENT ----------
def client():
Wait for server to start
time.sleep(1)
Create TCP socket
client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
Connect to server
client_socket.connect((HOST, PORT))
print("\nCLIENT: Connected to server")
print("Type 'exit' to terminate.\n")
while True:
Get message from user
message = input("Enter message: ")
Send message
client_socket.send(message.encode())
Receive response
response = client_socket.recv(1024).decode()
print("Server response:", response)
Exit condition
if message.lower() == "exit":
break
Close client socket
client_socket.close()
print("CLIENT: Disconnected.")
---------- MAIN PROGRAM ----------
Start server using a separate thread
server_thread = threading.Thread(target=server)
Start the server
server_thread.start()
Start the client
client()
36°C
Mostly sunny
Search

OUTPUT:


IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AM
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Krishna Priya P/AppData/Local/Programs/Python/Python314/.py
SERVER: Waiting for client connection...
SERVER: Connected to
CLIENT: Connected to server
('127.0.0.1', 56945)Type 'exit' to terminate.
Enter message: HELLO
SERVER: Client says -> HELLO
Server response: Server received: HELLO
Enter message: HOW ARE YOU?
SERVER: Client says -> HOW ARE YOU?
Server response: Server received: HOW ARE YOU?
Enter message: EXIT
SERVER: Client says -> EXIT
Server response:SERVER: Stopped.
Connection closed by server.
CLIENT: Disconnected.
>>>